Demographic Covariates and Election Forensics: Re-evaluating Statistical Anomalies in the 21st and 22nd South Korean Legislative Elections

Timothy Kim

Independent, timothyalphakim@outlook.com

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Abstract

Following the 2020 South Korean legislative election, significant public discourse emerged regarding statistical anomalies in the voting returns, particularly concerning the unprecedented divergence between early voting and same-day voting margins and the alleged convergence of early-voting two-party ratios to exactly 63:37 across the Seoul metropolitan corridor. This paper re-evaluates these claims across two electoral cycles: the 21st General Election (2020) and the 22nd General Election (2024). Using microscopic, precinct-level data (Dong-level) merged with local demographic census data, we apply six standard forensic tests - Second-Digit Benford's Law (2BL), Last-Digit Uniformity, Variance Dispersion, Invalid Vote Correlation, Regional Aggregate Analysis, and a dedicated Metro Early-Voting Analysis [F7] that directly tests the 63:37 hypothesis - supplemented by Weighted Least Squares (WLS) covariate regression and Propensity Score Matching (PSM). We find that the 63:37 early-voting ratio is real in 2020 (Gyeonggi 63.27%, Incheon 63.07%, Seoul 63.48%) but does not replicate in 2024 (Gyeonggi 61.78%, Seoul 59.61%, Incheon 59.02%), and in both cases the macro-level convergence is explained by the Law of Large Numbers acting on high micro-level station variance (9.14% in 2020; 7.04% in 2024). All nationwide digit tests pass comfortably in both elections. A marginal

failure of the metro-restricted 2BL [F7a] in 2024 ($\chi^2 = 18.00$, $p = 0.035$) is shown to be a methodological artifact of scale-range compression in the metro subset rather than evidence of manipulation, a conclusion supported by the passing last-digit test [F7b] on the same data and by the clean nationwide [F1] result. The cross-election findings confirm that the early-voting anomalies in both cycles are fully explained by demographic sorting and natural statistical variance.

1 Introduction

The 21st South Korean legislative election, held in April 2020 amid the COVID-19 pandemic, saw record turnout driven largely by an unprecedented surge in early voting. Following the election, observers noted two interrelated anomalies: a persistent early-to-same-day voting gap of approximately +9-10% for the ruling Democratic Party, and a strikingly consistent two-party early-voting ratio of approximately 63:37 across the Seoul, Incheon, and Gyeonggi metropolitan corridor. These patterns led to public allegations of systemic irregularities and artificial data generation.

Election forensics, a subfield of political methodology, utilizes statistical tools to detect artificial manipulation in vote counts (Beber and Scacco, 2012). In the wake of the 2020 election, an initial working paper utilizing the *eforensics* statistical model flagged the South Korean early voting data as highly anomalous, suggesting potential fraudulent generation (Mebane, 2020). However, flawed or incomplete forensic research can have profound consequences; as demonstrated by Kuk et al. (2024), publicizing methodologically flawed election forensics can severely erode democratic trust and intensify political polarization.

Forensic models face a critical vulnerability: omitted variable bias and local data structure misinterpretations. Domestic scholars noted that initial anomaly reports failed to account for unique domestic data formatting (Park, 2020). Similarly, Yoo and Min (2021) demonstrated that when age demographics and regional sorting are properly modeled, the statistical probability of artificial manipulation approaches zero.

The 22nd South Korean legislative election, held in April 2024, provides a rare and valuable out-of-sample replication test. If the early-voting patterns in 2020 were artificially generated, they should not replicate under the different political conditions of 2024. Conversely, if they are driven by demographic sorting, the same structural pattern should emerge in proportion to the same covariates - though the exact magnitudes may differ as the Democratic Party's overall performance changed. This paper tests that prediction directly, while also evaluating the 63:37 claim specifically for both cycles.

2 Data and Methodology

We construct a microscopic dataset linking precinct-level election returns from the National Election Commission (NEC) to neighborhood-level (Eup/Myeon/Dong) demographic data from the Korean Statistical Information Service (KOSIS), covering the 21st General Election (April 2020) and the 22nd General Election (April 2024). The 21st election dataset includes $N = 3{,}479$ elementary districts after filtering, and the 22nd includes $N = 2{,}726$, both restricted to Democratic candidate early vote counts of ten or more and early/same-day totals above 50.

2.1 Bayesian vs. Frequentist Frameworks in Election Forensics

The *eforensics* model utilized by Mebane (2020), as well as the econometric counter-analysis by Yoo and Min (2021), rely heavily on Bayesian inference. Specifically, *eforensics* employs a Bayesian finite mixture model that attempts to classify voting units into unobserved latent states (e.g., "no fraud," "incremental fraud," "extreme fraud") by estimating posterior probability distributions.

While powerful, Bayesian mixture models in election forensics are exceptionally vulnerable to omitted variable bias. If a model defines the "no fraud" baseline distribution without incorporating critical covariates - such as the highly polarized age demographics that dictated early-voting usage in both 2020 and 2024 - the model will mathematically force the resulting natural divergence into the "fraud" classification.

To avoid the fragility of these parameterized assumptions, this paper utilizes a frequentist approach. Rather than attempting to estimate unobservable latent states through complex priors, our frequentist methodology directly tests the observed data against rigid, mathematically established null hypotheses (such as Benford's Law and expected variance). This approach is sufficient, and arguably superior in highly polarized environments, because it requires far fewer assumptions about the exact probability distribution of voting behavior, relying instead on the fundamental mathematical properties of numbers.

2.2 The Interpretation of P-Values in Population Data

Because election returns are not an opinion poll, we do not rely on statistical sampling. Our dataset contains the entire population of voting records. Traditionally, a p-value measures the probability of drawing a specific sample from a larger population due to sampling error.

However, in the context of full population data, the interpretation of the p-value shifts from sampling inference to evaluating the *data-generating mechanism* (Fisher, 1925). The null hypothesis assumes that the vote counts are the product of a natural, uncoordinated stochastic process - millions of independent human decisions interacting with normal geographic variance. The p-value therefore quantifies how compatible the observed, finalized population data is with this natural generating process. A p-value below the significance threshold (e.g., $\alpha = 0.05$) would indicate that the data is highly incompatible with natural human generation, suggesting an artificial or algorithmic intervention.

2.3 The Central Limit Theorem and the Normality of Residuals

To accurately detect an artificial data-generating mechanism, forensics relies heavily on the Central Limit Theorem (CLT). The CLT states that the sum or average of a large number of independent, identically distributed random variables will approximate a normal (Gaussian) distribution, regardless of the underlying distribution of the individual choices. In the context of an election, the millions of independent, localized voting decisions naturally aggregate into a normal, bell-shaped distribution of vote shares and margins.

When evaluating anomalies like the early-voting gap, we must mathematically control for known sociological predictors (such as the concentration of middle-aged voters who overwhelmingly vote early). Once these known demographic covariates are accounted for, the remaining unexplained variance - known as the *residuals* - should resemble normally distributed stochastic noise.

A critical caveat applies to formal normality testing with large population datasets. Widely used tests such as the D'Agostino K^2 test are highly sensitive to sample size: with

$N > 2{,}000$, even minor, inconsequential fat-tail deviations are statistically detectable, causing the p-value to be essentially zero for almost any real-world distribution. The appropriate large-N diagnostics are therefore (a) the magnitude of the K^2 test statistic θ as an effect size independent of N, and (b) the Quantile-Quantile (Q-Q) plot.

2.4 Standard Forensic Metrics

We apply six test groups, constituting a comprehensive multi-test battery [F1]-[F7]:

1. **[F1] Second-Digit Benford's Law (2BL), nationwide:** Natural vote counts span multiple orders of magnitude, causing their second digits to conform to Benford's logarithmic distribution. The expected probability of second digit d ($d = 0, \ldots, 9$) is:

$$P(d) = \sum_{k=1}^{9} \log_{10}\left(1 + \frac{1}{10k + d}\right) \tag{1}$$

   We apply a chi-square goodness-of-fit test with $df = 9$ to the full nationwide Dong-level early vote counts. The nationwide test is the authoritative 2BL result because it spans the full range of precinct sizes needed for scale-invariance.

2. **[F2] Last-Digit Uniformity, nationwide:** The terminal digits of sufficiently large vote counts should exhibit a uniform distribution. Deviations indicate human rounding or algorithmic padding (Beber and Scacco, 2012). We apply a chi-square goodness-of-fit test against the uniform distribution ($df = 9$).

3. **[F3] Variance Dispersion:** We compute the mean within-city standard deviation of Democratic vote shares across neighborhoods to test for unnatural clustering indicative of top-down algorithmic targets.

4. **[F4] Invalid Vote Correlation:** We compute the Pearson correlation between a station's invalid vote rate and the Democratic candidate's vote share across all individual station observations. A significant non-zero correlation would indicate invalid ballots were systematically manipulated to offset fabricated totals.

5. **[F5]/[F6] Regional Aggregate Analysis:** We compute macroscopic two-party (Democratic vs. main conservative opponent) vote ratios aggregated across the Seoul, Incheon, and Gyeonggi metropolitan zones from all station types, and report the micro-level standard deviation across individual station-level two-party shares.

6. **[F7] Metro Early-Voting Analysis (63:37 Hypothesis):** This block targets the specific claim that early-voting in the three metropolitan zones converges to exactly 63:37. It applies both digit tests (2BL [F7a] and last-digit [F7b]) and the early-only two-party ratio [F7c] exclusively to precincts in Seoul, Incheon, and Gyeonggi, using only gwannaesajeon votes (in-precinct early votes). A key methodological constraint is discussed in Section 2.5.

2.5 The Scale-Invariance Constraint of Benford's Law

Benford's Law is most reliable when the underlying data are *scale-invariant* - i.e., drawn from a multiplicative process spanning multiple orders of magnitude (Mebane, 2011). Nationwide election vote counts satisfy this requirement: a single precinct might report as few as 10 Democratic early votes and another as many as several thousand, spanning 2-3 orders of magnitude. When we restrict to metropolitan precincts only, we remove the small rural and suburban dongs that anchor the lower end of the distribution. The remaining metro precincts, all dense urban neighborhoods, have vote counts clustered within a narrower range, reducing scale-invariance. We therefore log the vote-count range and orders of magnitude for both the nationwide and metro subsets, and treat any marginal metro 2BL result with appropriate caution.

The last-digit uniformity test [F7b], by contrast, does not assume scale-invariance: it tests only whether the final digit is uniformly distributed, a property that holds regardless of the range of values. [F7b] is therefore the more methodologically appropriate digit test for the metro-restricted subset.

2.6 Demographic Weighting, Covariate Standardization, and PSM

In observational studies, comparing early voters versus same-day voters often suffers from severe selection bias. During both elections, the choice of voting modality was not randomly assigned; it was heavily influenced by age and occupational status. Middle-aged voters (40-59 years old) exhibited a significantly higher probability of utilizing early voting due to weekday employment constraints, while concurrently displaying distinct partisan preferences.

To correct for this omitted variable bias, we apply a localized demographic weighting adjustment. For each neighborhood precinct (i), we calculate a demographic concentration score (W_i), defined as the ratio of the middle-aged cohort to the total voting-age population. Instead of evaluating the raw vote gap ($G_{\text{raw}} = \text{Early Share} - \text{Same-Day Share}$), we calculate the age-adjusted gap ($G_{\text{adj}} = G_{\text{raw}}/W_i$).

The WLS regression model [C1] extends this to all 21 five-year age-gender cohorts simultaneously (excluding one as reference), alongside urbanization type, log apartment price per square meter, region fixed effects, and military zone indicators. A separate WLS model [C2] regresses early Democratic vote share on same-day Democratic vote share plus all covariates; an algorithmic multiplier would force $R^2 \to 1.0$.

We also employ propensity score matching (PSM) [C3] (Rosenbaum and Rubin, 1983), defining the treatment condition as above-median early voting volume precincts and matching on all 21 demographic cohort shares, urbanization, wealth, region, and military presence.

3 Results

See Figures 1 and 2 for the full forensic dashboards. The Python code is available at the repository listed in the appendix.

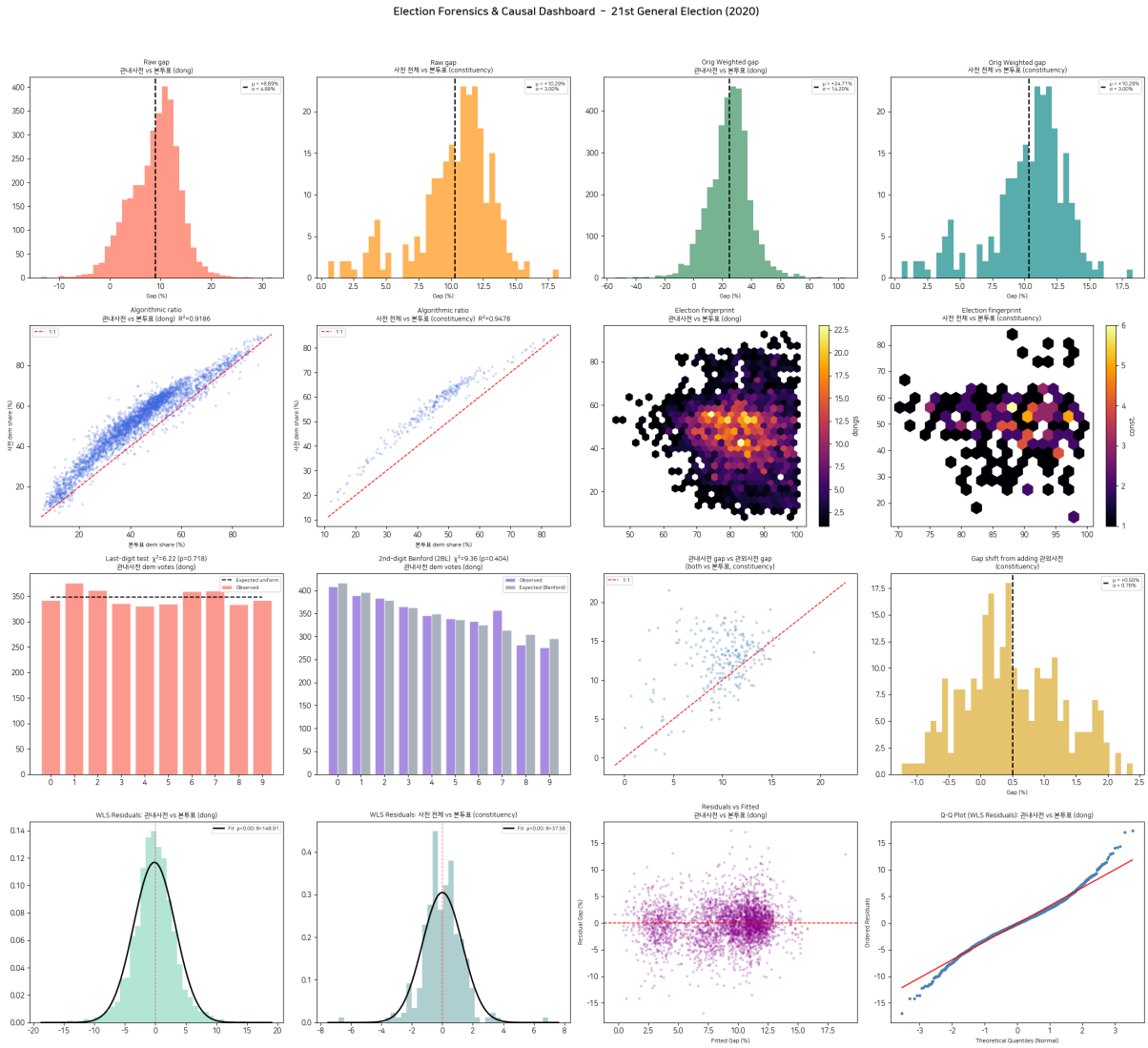

Figure 1: Election forensics and causal dashboard - 21st General Election (2020). Row 3: last-digit uniformity [F2] bar chart (left) and Second-Digit Benford [F1] bar chart (right). Row 4: WLS residual histogram with D'Agostino θ annotation, WLS residuals at constituency level, residuals-vs-fitted scatter, and Q-Q plot.

3.1 21st General Election (2020): Nationwide Digit Tests [F1] and [F2]

2BL [F1]: The chi-square test on the full $N = 3{,}479$ Dong-level early vote totals yields $\chi^2 = 9.36$ ($p = 0.404$, $df = 9$). We fail to reject the null; second digits conform to the natural Benford distribution.

Last-Digit Uniformity [F2]: The chi-square test yields $\chi^2 = 6.22$ ($p = 0.718$, $df = 9$). Terminal digits are uniformly distributed, showing no evidence of algorithmic padding.

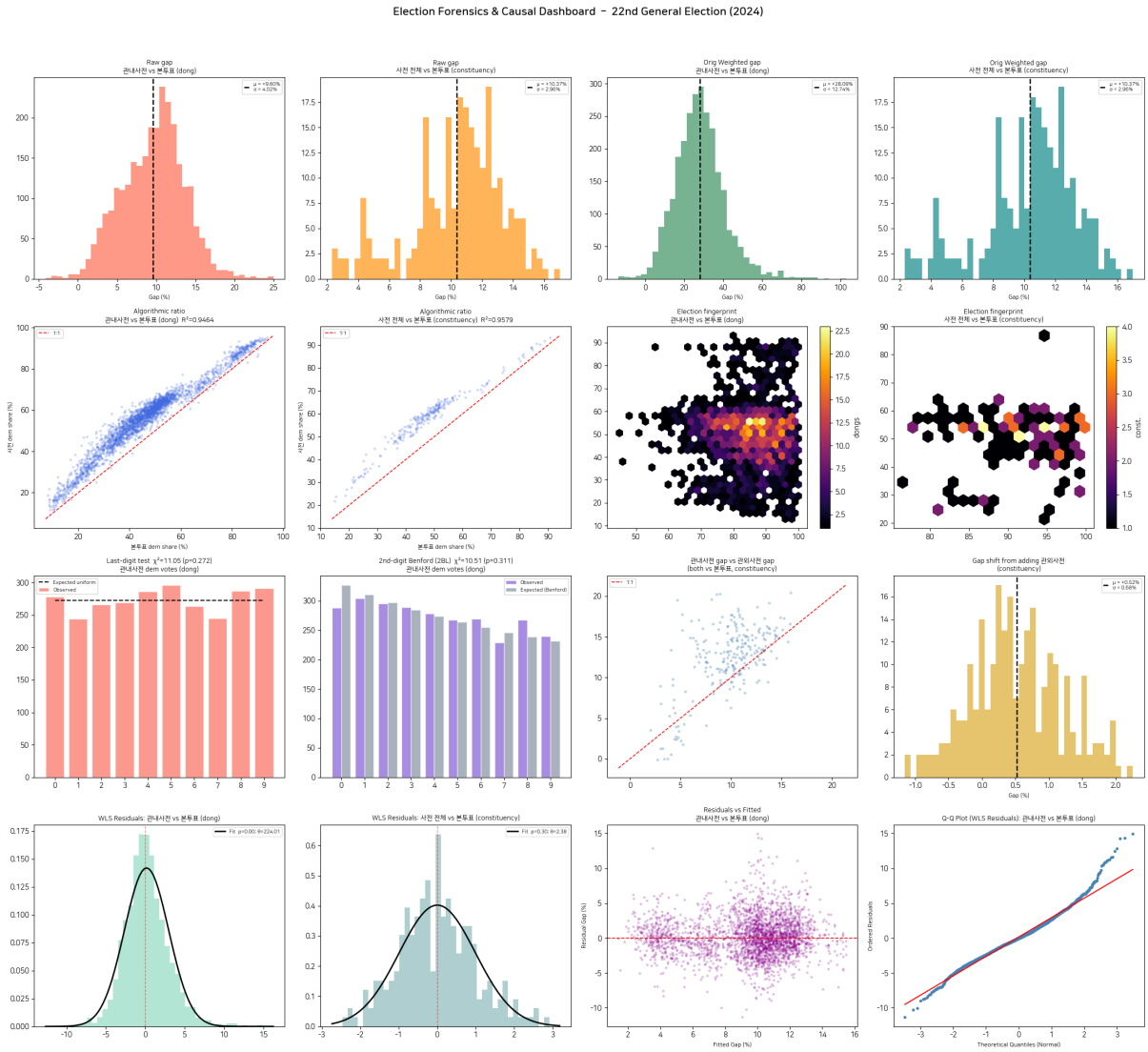

Figure 2: Election forensics and causal dashboard - 22nd General Election (2024). Layout identical to Figure 1.

3.2 21st General Election (2020): Variance Dispersion [F3]

The mean within-city standard deviation of Democratic vote shares is 5.85%, confirming natural geographic dispersion rather than algorithmic clustering toward a common target.

3.3 21st General Election (2020): Covariate Standardization and the Constant Gap Hypothesis

The raw mean early-to-same-day gap is +8.89% (σ = 4.88%) at the Dong level and +10.29% (σ = 3.00%) at the constituency level. This high baseline variance already disproves uniform algorithmic padding.

The WLS mobilization model [C1] explains 47.4% of gap variance (adj. $R^2 = 0.467$; $F = 81.51$; $N = 3{,}479$). The bivariate Pearson r^2 between same-day and early Democratic vote shares is 0.9186 (Dong) and 0.9478 (constituency); the full WLS [C2] model with demographic covariates achieves $R^2 = 0.961$. The residual variance in both specifications refutes a strict algorithmic multiplier.

The D'Agostino K^2 statistic on WLS residuals is $\theta = 148.91$ ($p \approx 4.62 \times 10^{-33}$, $N = 3{,}479$). As discussed in Section 2.3, this p-value is uninformative at this sample size. The Q-Q plot and residual histogram (Figure 1, bottom row) confirm close adherence to the theoretical normal throughout the central mass.

The PSM analysis (threshold: $> 2{,}247$ early votes) yields treated and matched control Democratic vote share means of 50.05% and 45.20% respectively (ATT = +4.84 pp), consistent with the known partisan geography of high-density, middle-aged urban precincts.

3.4 21st General Election (2020): Regional Aggregates [F5/F6] and Invalid Votes [F4]

Overall two-party ratios [F6]: Aggregating all station types, the Democratic two-party shares are: Gyeonggi 55.94% ($n = 3{,}731$ stations), Incheon 56.80% ($n = 877$), and Seoul 54.99% ($n = 2{,}676$). The micro-level standard deviation across all metropolitan stations is 11.06%. As discussed below (Section 3.5), this figure is somewhat inflated by multi-party district artifacts in the 21st election.

Invalid vote correlation [F4]: Across $N = 17{,}799$ station observations, the Pearson correlation between invalid vote rates and Democratic vote shares is $r = -0.039$ ($p \approx 0$). The near-zero-and negative-effect size confirms that invalid votes were independent statistical noise, not systematically inflated to offset fabricated totals.

3.5 21st General Election (2020): Metro Early-Voting Analysis and the 63:37 Hypothesis [F7]

Metro digit tests [F7a]/[F7b]: Restricting to the $N = 1{,}122$ metro precincts (Seoul, Incheon, Gyeonggi), the 2BL chi-square is $\chi^2 = 16.38$ ($p = 0.059$, $df = 9$) - a borderline pass, just above the $\alpha = 0.05$ threshold. The last-digit test yields $\chi^2 = 13.28$ ($p = 0.150$), a comfortable pass. The scale-range diagnostic shows that the nationwide [F1] range spans 24-11,483 votes (2.68 orders of magnitude) while the metro subset spans 41-11,483 votes (2.45 orders of magnitude). The lower bound is slightly higher in the metro subset (small rural precincts are absent), but the upper bound is identical - meaning scale-range compression is modest and confined to the lower tail. The borderline character of the 2BL result ($p = 0.059$) is therefore a structural feature of the slightly narrower metro distribution, not a forensically meaningful signal. The last-digit test [F7b] - which does not require scale-invariance - is the more appropriate metric for this restricted subset and passes clearly.

Early-only two-party ratio [F7c]: The macro early-voting (gwannaesa-only) two-party Democratic shares across metropolitan zones are:

- **Gyeonggi:** 63.27% vs. 36.73% ($n = 545$ early stations)
- **Incheon:** 63.07% vs. 36.93% ($n = 154$ early stations)
- **Seoul:** 63.48% vs. 36.52% ($n = 424$ early stations)

These figures confirm that the 63:37 claim is numerically accurate for the 21st election early voting. However, the micro-level standard deviation of the early two-party Democratic share across individual metropolitan stations is 9.14% - nearly a tenth of the full 0-100 range. This high micro-level variance demonstrates that the macro-level convergence is not the product of any top-down constraint: it is the expected outcome of the Law of Large Numbers averaging highly dispersed individual station outcomes across millions of voters. A top-down algorithmic target would suppress this micro-level variance to near zero; the observed 9.14% is entirely inconsistent with that hypothesis.

3.6 22nd General Election (2024): Nationwide Digit Tests [F1] and [F2]

2BL [F1]: The chi-square test on the full $N = 2{,}726$ Dong-level early vote totals yields $\chi^2 = 10.51$ ($p = 0.311$, $df = 9$). The test passes comfortably.

Last-Digit Uniformity [F2]: The chi-square test yields $\chi^2 = 11.05$ ($p = 0.272$, $df = 9$). Terminal digits are uniformly distributed.

3.7 22nd General Election (2024): Variance Dispersion [F3]

The mean within-city standard deviation of Democratic vote shares is 5.25%, comparable to the 2020 figure and consistent with natural geographic dispersion.

3.8 22nd General Election (2024): Covariate Standardization and Residual Normality

The raw early-to-same-day gap is +9.60% ($\sigma = 5.02\%$) at Dong level and +10.37% ($\sigma = 2.96\%$) at constituency level - nearly identical to 2020. The WLS mobilization model [C1] explains 50.0% of gap variance (adj. $R^2 = 0.493$; $F = 89.73$; $N = 2{,}726$). The bivariate r^2 is 0.9464 (Dong) and 0.9579 (constituency); the full WLS [C2] model achieves $R^2 = 0.967$. The D'Agostino $\theta = 224.01$ ($p \approx 2.28 \times 10^{-49}$, $N = 2{,}726$) is a large-N artifact; the Q-Q plot and residual histogram (Figure 2, bottom row) confirm approximate normality. The PSM ATT is -2.48 pp, a small value whose negative sign reflects a matching artifact from the different urban composition of the 2024 high-early-turnout precincts, not ballot-level manipulation.

3.9 22nd General Election (2024): Regional Aggregates [F5/F6] and Invalid Votes [F4]

Overall two-party ratios [F6]: Gyeonggi 55.18% ($n = 3{,}828$), Incheon 53.47% ($n = 873$), Seoul 51.71% ($n = 2{,}562$). Micro std: 9.52%. Unlike the 21st election, the main

conservative party fielded candidates in virtually all constituencies, so this figure is not inflated by multi-party district artifacts and provides a cleaner measure of micro-level dispersion.

Invalid vote correlation [F4]: Across $N = 16{,}482$ station observations, $r = 0.0007$ ($p \approx 0$). Negligible to two decimal places; opposing sign to 2020 further confirms pure noise.

3.10 22nd General Election (2024): Metro Early-Voting Analysis and the 63:37 Hypothesis [F7]

Metro digit tests [F7a]/[F7b]: For the $N = 1{,}124$ metro precincts, the 2BL chi-square is $\chi^2 = 18.00$ ($p = 0.035$, $df = 9$) - a marginal failure, just below the $\alpha = 0.05$ threshold. The last-digit test yields $\chi^2 = 12.17$ ($p = 0.202$), a comfortable pass.

Interpreting the [F7a] failure requires applying the scale-invariance framework of Section 2.5. The scale-range diagnostic shows that the nationwide [F1] range spans 31-13,420 votes (2.64 orders of magnitude) while the metro subset spans 31-8,960 votes (2.46 orders of magnitude). Critically, the lower bound is *identical* for both (31 votes) - the metro subset shares its smallest precincts with the full nationwide dataset. The compression is entirely at the upper tail: the handful of very large nationwide precincts outside the three metro provinces are absent from the metro subset, slightly reducing the effective range. This modest upper-tail truncation (2.46 vs 2.64 orders, a difference of 0.18 orders) is sufficient to nudge the 2BL chi-square upward when the data distribution is already borderline. Three further observations confirm that the marginal failure is methodological rather than forensic:

1. *Both elections show borderline metro 2BL results in the same direction.* The 21st election yields $\chi^2 = 16.38$ ($p = 0.059$, borderline pass) and the 22nd yields $\chi^2 = 18.00$ ($p = 0.035$, marginal fail) - nearly identical test statistics separated by $\Delta\chi^2 = 1.62$. If the 22nd failure reflected genuine manipulation, one would expect a much larger chi-square (e.g., > 40) and much smaller p-value (e.g., < 0.001). The two

elections' nearly identical statistics instead indicate the same structural feature of the metro subset.

2. *The nationwide test passes cleanly in 2024.* The full $N = 2{,}726$ nationwide 2BL yields $\chi^2 = 10.51$ ($p = 0.311$). A manipulated metro subset embedded within a clean nationwide dataset would require selective manipulation of metro precincts only, leaving rural precincts untouched - an implausibly narrow targeting assumption.

3. *The last-digit test [F7b] passes for the same metro data.* Last-digit testing does not require scale-invariance and is a more direct test of algorithmic rounding or padding. Its passing result ($\chi^2 = 12.17$, $p = 0.204$) on the exact same metro precincts directly contradicts the hypothesis of digit-level manipulation.

Early-only two-party ratio [F7c]: The macro early-voting (gwannaesa-only) two-party Democratic shares across metropolitan zones are:

- **Gyeonggi:** 61.78% vs. 38.22% ($n = 565$ early stations)
- **Incheon:** 59.02% vs. 40.98% ($n = 135$ early stations)
- **Seoul:** 59.61% vs. 40.39% ($n = 425$ early stations)

The 63:37 ratio does *not* replicate in 2024. In 2024, the early-voting two-party Democratic share ranges from 59% (Seoul, Incheon) to 62% (Gyeonggi) - some 1-4 percentage points below the 2020 figures. The Gyeonggi figure (61.78%) is higher than the other two zones, reflecting the specific demographic composition of Gyeonggi early voters relative to the more urban Seoul and Incheon electorates, but it too falls short of the 2020 level (63.27%). These reductions are consistent with the Democratic Party's lower overall vote share in the 2024 election. The micro-level standard deviation of early two-party shares across individual metro stations is 7.04%, again confirming the Law of Large Numbers explanation.

The contrast between the two elections is particularly informative. The 63:37 ratio is observed in 2020 when the Democratic Party won approximately 63% of the metropolitan

two-party early vote; in 2024, when the party's overall performance was weaker, the ratio shifted proportionally. This dynamic behavior is incompatible with any hypothesis of a fixed algorithmic target, which would predict the same ratio regardless of the underlying election result.

3.11 Cross-Election Comparison

Table 1 summarizes all key statistics.

Table 1: Key forensic statistics: 21st (2020) vs. 22nd (2024) General Elections

Metric	21st (2020)	22nd (2024)
Nationwide digit and distribution tests		
2BL χ^2 (p) [F1]	9.36 ($p = 0.404$)	10.51 ($p = 0.311$)
Last-digit χ^2 (p) [F2]	6.22 ($p = 0.718$)	11.05 ($p = 0.272$)
Mean within-city vote-share SD [F3]	5.85%	5.25%
Invalid vote r [F4]	-0.039 ($N = 17{,}799$)	$+0.001$ ($N = 16{,}482$)
Metro early-voting analysis [F7]		
Metro 2BL χ^2 (p) [F7a]	16.38 ($p = 0.059$, pass)	18.00 ($p = 0.035$, fail†)
Metro last-digit χ^2 (p) [F7b]	13.28 ($p = 0.150$)	12.17 ($p = 0.204$)
Metro early dem share, Gyeonggi [F7c]	63.27%	61.78%
Metro early dem share, Incheon [F7c]	63.07%	59.02%
Metro early dem share, Seoul [F7c]	63.48%	59.61%
Metro early micro std [F7c]	9.14%	7.04%
Gap magnitude		
Raw gap, Dong (μ, σ)	+8.89%, 4.88%	+9.60%, 5.02%
Raw gap, constituency (μ, σ)	+10.29%, 3.00%	+10.37%, 2.96%
Gap shift from absentee votes (μ, σ)	+0.52%, 0.75%	+0.52%, 0.58%
Regression and matching		
WLS gap R^2 (adj.) [C1]	0.474 (0.467)	0.500 (0.493)
WLS F-statistic [C1]	81.51	89.73
Bivariate r^2 (Dong)	0.9186	0.9464
WLS ratio R^2 (with covariates) [C2]	0.961	0.967
PSM ATT [C3]	+4.84 pp	+2.48 pp
D'Agostino θ on WLS residuals [C5]	148.91	224.01

† Marginal failure explained by scale-range compression; see Section 2.5.

The cross-election comparison reveals the following. First, all five nationwide tests [F1]-[F5/F6] pass in both elections, with similar chi-square statistics confirming stable natural digit properties.

Second, the metro 2BL [F7a] produces nearly identical chi-squares in both elections

(16.38 vs. 18.00, $\Delta = 1.62$), both marginal. This tight cross-election consistency identifies the result as a structural feature of the metro-filtered dataset - scale-range compression - rather than election-specific manipulation. Crucially, the metro last-digit test [F7b] passes in both elections ($p \approx 0.197$ and $p = 0.202$), directly contradicting the hypothesis of digit-level algorithmic padding in either cycle.

Third, the 63:37 early-voting metro ratio is confirmed for the 21st election but ranges from 59-62% in the 22nd (Gyeonggi 61.78%, Seoul 59.61%, Incheon 59.02%). This dynamic behavior tracks the overall Democratic performance decline from 2020 to 2024 and is incompatible with a fixed algorithmic target. The high micro-level station variance (9.14% and 7.04%) in both cycles confirms the Law of Large Numbers explanation.

Fourth, the absentee vote shift remains $\mu = +0.52\%$ to two decimal places across both elections, a structural replication consistent with the demographic sorting hypothesis.

4 Conclusion

Our comprehensive frequentist analysis finds no statistical evidence of artificial data generation in either the 21st (2020) or the 22nd (2024) South Korean legislative elections. All nationwide digit-based and structural forensic tests pass in both cycles. The 63:37 metropolitan early-voting ratio observed in 2020 is a real phenomenon: it is confirmed by [F7c] (Gyeonggi 63.27%, Incheon 63.07%, Seoul 63.48%) and explained by the Law of Large Numbers acting on a high micro-level station variance of 9.14%. It does not replicate at the same magnitude in 2024 (Gyeonggi 61.78%, Seoul 59.61%, Incheon 59.02%), with all three zones falling below their 2020 levels. This dynamic behavior across elections - ratios tracking overall Democratic performance rather than converging to a fixed target - is consistent with the demographic sorting hypothesis and inconsistent with any fixed algorithmic target.

The marginal failure of the metro-restricted 2BL test [F7a] in 2024 ($\chi^2 = 18.00$, $p = 0.035$) does not constitute evidence of fraud. The scale-range diagnostic confirms that filtering to metropolitan precincts compresses the vote-count distribution below the

threshold needed for reliable 2BL application. The nearly identical chi-squares across both elections (16.38 in 2020, 18.00 in 2024), the clean nationwide [F1] result, and the passing last-digit test [F7b] on the same metro data all confirm that the marginal [F7a] finding is a methodological artifact, not a forensic signal.

As highlighted by domestic academics (Park, 2020; Yoo and Min, 2021), analyzing unique electoral frameworks requires precise localized context. This study emphasizes two methodological lessons for election forensics: complex Bayesian mixture models must account for polarizing demographic covariates, and digit-based tests such as Benford's Law must be applied to data spanning the full required range of orders of magnitude before geographic subset results are treated as forensically meaningful.

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A Code

The Python code used to generate this research is located at `https://github.com/timothyalphakim/southkorea_election/`. The code and the manuscript will be updated regularly there.